

Summary This paper studies a deep linear residual architecture, where there are n layers in each residual block, and D/n total blocks: $f(x) = \prod_{i=1}^{D/n} (I + \prod_{j=1}^n W_{i,j})$. In contrast, prior theoretical work (Hardt & Ma 2016) only studied the deep linear case where $n = 1$, which is (in the non-linear setting) does not perform as well as higher n in practice.

To understand the stability of gradient descent in this architecture, the authors consider the maximum eigenvalue of the Hessian obtained at solution obtained via gradient descent. Under extremely strong conditions on the initialization of the network, the authors show that that top eigenvalue scales like $D\rho^{2-2n/D}(\rho^{n/D} - 1)^{2-2/n}$, where ρ is the spectral norm of the ground truth linear map. They observe that as long as $\rho > 0$, this value is smaller for any $n > 1$ than for $n = 1$. Note that it is not necessarily monotone though: in some settings it is larger for $n = 3$ than $n = 2$.

The authors provide some simulations showing that the convergence is faster for $n > 1$.

Strengths Since $n = 1$ performs poorly in practice, understanding the mechanism behind this is interesting.

Weaknesses

1. The assumptions on the result are incredibly strong to the point that I don't believe this paper provides any technical value, especially because the setting is already a simple linear architecture. Namely, in Theorem 7 (the basis for their results) the authors assume that the matrices are initialized to be orthogonalizable in the same basis as that ground truth map. These assumptions are not standard for studying deep linear nets
2. This papers motivates themselves by comparing to (Hardt & Ma 2016) which studied the case of $n = 1$, but there is no explanation or comparison of their results. On the whole, discussion of related work is lacking.
3. The paper makes broad conclusions based on weak evidence. For example, they argue at the bottom of page 10 that "the reason why residual networks show good training and generalization" is the multiple layers per block. However, it is not clear what (if any) insights this paper gives on generalization. If the authors want to claim something about optimization, they should at least state a formal corollary of their result about the step size and optimization time.